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Probability, Bayes and medical decision making

Week 4: Why a 99 per cent accurate test can still be wrong most of the time it fires.

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By the end of this week you should be able to

Learning outcomes served: LO2, LO5

- Apply Bayes' rule to convert between sensitivity and positive predictive value.
- Explain the effect of prevalence on PPV and NPV while sensitivity and specificity are held fixed.
- Read a confusion matrix and derive every standard metric from it.
- Choose an operating threshold according to the relative cost of the two error types.
- Explain why screening an unselected population is a different problem from testing a symptomatic one.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- Every diagnostic test has two error rates and operates in a population with some prevalence. Clinicians, patients and, frequently, machine-learning practitioners confuse the probability of a positive test given disease with the probability of disease given a positive test.

- These are different numbers, sometimes by an order of magnitude, and Bayes' rule is what converts between them.

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The method

Part 2 · Concepts, then the mathematics

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Conditional probability

P(A|B) is the probability of A given that B has occurred. P(positive test | disease) and P(disease | positive test) are not the same quantity.

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Bayes' rule

P(A|B) = P(B|A)P(A) / P(B). It reweights the likelihood by the prior. In diagnosis the prior is the prevalence.

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Key concepts

Sensitivity P(test positive | disease present). The true positive rate, also called recall.

Specificity P(test negative | disease absent). The true negative rate.

Prevalence The proportion of the tested population that has the condition. It is a property of the population, not of the test, and it drives PPV.

Positive predictive value P(disease | test positive). This is what the patient in front of you actually wants to know, and it collapses as prevalence falls.

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Key concepts (continued)

Negative predictive value P(no disease | test negative). Usually high when prevalence is low, which is why a negative result is often more informative in screening.

The base-rate fallacy Judging a positive result by the test's accuracy while ignoring how rare the condition is. It is the single most consequential probabilistic error in clinical machine learning.

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Bayes' rule

$$P(A \mid B) = P(B \mid A) \cdot P(A) \;/\; P(B)$$

Converts between the two conditional probabilities that are routinely confused in diagnosis.

P(A) the prior: prevalence, before the test

P(B|A) the likelihood: sensitivity, if A is disease and B a positive test

P(A|B) the posterior: the positive predictive value

P(B) the evidence, which normalises the result

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Positive predictive value, written out

$$PPV = (Se \cdot p) / [\, Se \cdot p + (1 - Sp) \cdot (1 - p) \,]$$

Sensitivity and specificity belong to the test. Prevalence belongs to the population you chose to test. PPV depends on both, which is why the same test behaves differently in screening and in clinic.

Se sensitivity, P(test positive | disease)

Sp specificity, P(test negative | no disease)

p prevalence in the tested population

NPV = Sp(1−p) / [Sp(1−p) + (1−Se)p]

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A 99% sensitive, 99% specific test, at three prevalences

The test never changed. The population did.

Prevalence	Tested positive	Truly ill	PPV
0.1% (screening)	1,098 of 100,000	99	9.0%
1%	1,089 of 100,000	990	50.0%
10% (symptomatic clinic)	10,800 of 100,000	9,900	91.7%
37.3% (the WDBC cohort)	37,300 of 100,000	36,927	98.3%

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Reading a confusion matrix

Every one of these is derivable from four counts.

Metric	From the matrix	What it answers
Sensitivity / recall	TP / (TP + FN)	Of those who are ill, how many are found?
Specificity	TN / (TN + FP)	Of those who are well, how many are cleared?
Precision / PPV	TP / (TP + FP)	Of those flagged, how many really are ill?
NPV	TN / (TN + FN)	Of those cleared, how many really are well?
F1	2·P·R / (P + R)	One number when precision and recall matter equally
Accuracy	(TP + TN) / n	Almost useless when the classes are imbalanced

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Worked example

Part 3 · A very good test in a low-prevalence population

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A very good test in a low-prevalence population

- Take a test with 99 per cent sensitivity and 99 per cent specificity, applied to a condition with a prevalence of 1 in 1,000. Imagine 100,000 people.
- 100 of them have the condition, and the test correctly identifies 99 of those. 99,900 do not have it, and the test wrongly flags 1 per cent of them, which is 999 people. So 1,098 people test positive, and only 99 of them are ill.
- The positive predictive value is 99/1098, which is about 9 per cent. A test that is right 99 per cent of the time produces a positive result that is wrong more than nine times out of ten. Nothing is broken; this is what those numbers mean.
- Now raise the prevalence to 10 per cent, as it might be in a symptomatic clinic rather than an unselected screening population. Of 100,000 people, 10,000 are ill and 9,900 are correctly flagged; 90,000 are well and 900 are wrongly flagged. PPV rises to 9900/10800, about 92 per cent.
- The test did not change. The population did.

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Sensitivity and specificity belong to the test. Predictive value belongs to the test and the population together.

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Try this interactively

Medical Test Calculator

Why can a 99 per cent accurate test be wrong most of the times it fires?

`weeks/week04.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · Calibration is where this reappears

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Calibration is where this reappears

- Modern models, including fine-tuned foundation models, frequently produce confident-looking probabilities that are not calibrated: a set of predictions at "0.9" does not contain 90 per cent positives. Everything in this week depends on the input probabilities meaning what they claim.
- Week 12 returns to calibration curves and why a well-discriminating model can still be badly calibrated.

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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A screening model is reported as "94 per cent accurate" on a cohort where 6 per cent of participants have the condition. Before reading any further, what accuracy would a model that always predicts "no disease" achieve on that cohort, and what does that tell you about the reported figure?

Think like a scientist. Bring your answer to the laboratory.

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From confusion matrix to clinical consequence

This week's laboratory · Move a decision threshold and watch the clinical meaning of the model change. · 2 hours

- Fit a logistic regression and obtain predicted probabilities on a held-out set. Produce the confusion matrix at the default 0.5 threshold.
- From that matrix compute sensitivity, specificity, PPV, NPV, precision, recall and F1 by hand, using numpy rather than sklearn, so that you can see where each comes from.
- Sweep the threshold from 0.05 to 0.95 and plot sensitivity and specificity against it. Mark the point where they cross.
- Plot the ROC curve and compute the AUC. Then plot the precision-recall curve. Explain which is more informative for a rare condition and why.
- Using the Bayes calculator from the lecture, take your model's measured sensitivity and specificity and compute PPV at prevalences of 0.37 (the dataset), 0.10 and 0.005 (population screening).
- Choose a threshold and defend it. State whose interests you weighted and what a false negative costs relative to a false positive in your chosen setting.

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Self-check

- Prevalence falls from 10 per cent to 1 per cent while sensitivity and specificity are unchanged. What happens to PPV?
- A) It rises
- B) It falls substantially
- C) It is unchanged
- D) It depends on sample size

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Self-check

- A model has recall 0.95 and precision 0.30 for a rare disease. What is the practical consequence?
- A) It misses most cases
- B) It catches almost all cases, but around 70 per cent of its alarms are false
- C) It is well calibrated
- D) It is unusable

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References and further reading

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Questions

Next week: continue to Week 5.

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